

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0088

Aspen 345 kV
August 18, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0088. The Interconnection Customer proposes a 786.0 MW natural gas combined cycle facility interconnecting at the Aspen 345 kV substation in the Virginia area. The request was received on December 19, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 69 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,810,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Westchase – Coffeen N 138 KV is overloaded to 139.2% of its 216 MVA emergency rating for the loss of the Crandall Substation – Dayton 345 KV. The project contributes 5.5% DFAX (43.4 MW) to this overload.

The Coffeen N – Harrisonburg 138 KV is overloaded to 139.0% of its 217 MVA emergency rating for the loss of the Dayton – Crandall 345 KV Line. The project contributes 5.5% DFAX (43.4 MW) to this overload.

For the loss of Dayton Substation – Crandall 345 KV Line, the Station 54 – Westchase 138 KV line loads to 135.9% of its 236 MVA rating. Project DFAX: 5.5%.

Dayton – Harrisonburg 138 KV Line: 109.7% loading (299 MVA) following the outage of the Dayton – Chatom 345 KV; DFAX = 16.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Coffeen N – Westchase 138 KV	Rebuild line with higher-capacity conductor	\$5,337,000	45% (\$2,381,000)
Coffeen N – Harrisonburg 138 KV	Rebuild line with higher-capacity conductor	\$891,000	45% (\$398,000)
Sta 54 – Westchase 138 KV Line	Rebuild line with higher-capacity conductor	\$3,245,000	45% (\$1,448,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dayton – Harrisonburg 138 KV Line	Reconductor line	\$583,000	100% (\$583,000)
	Total allocated to project		\$4,810,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.